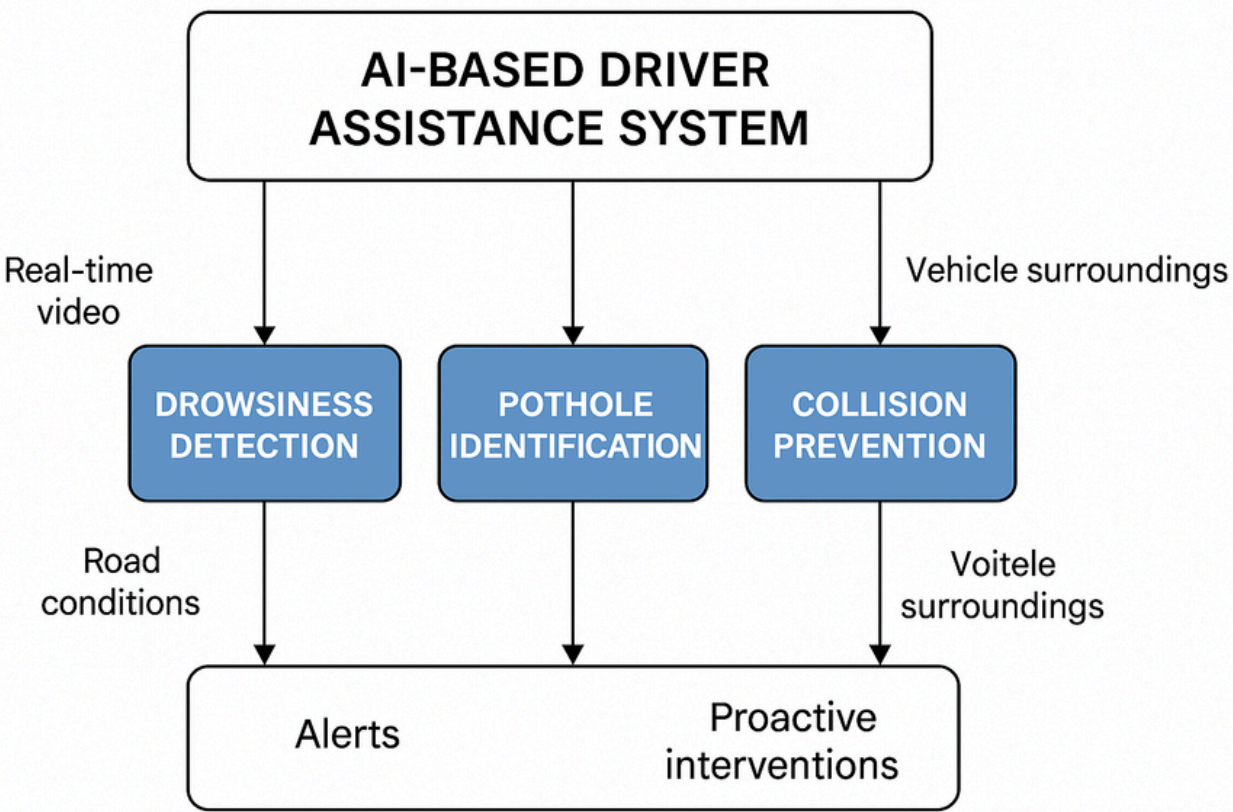


Project Title: Drive Safe using AI Assistance

Problem Statement:

Develop an AI-based Driver Assistance System that integrates drowsiness detection, pothole identification, and collision prevention to enhance road safety. The system aims to reduce accidents by providing real-time alerts and proactive interventions for drivers.

Architecture:



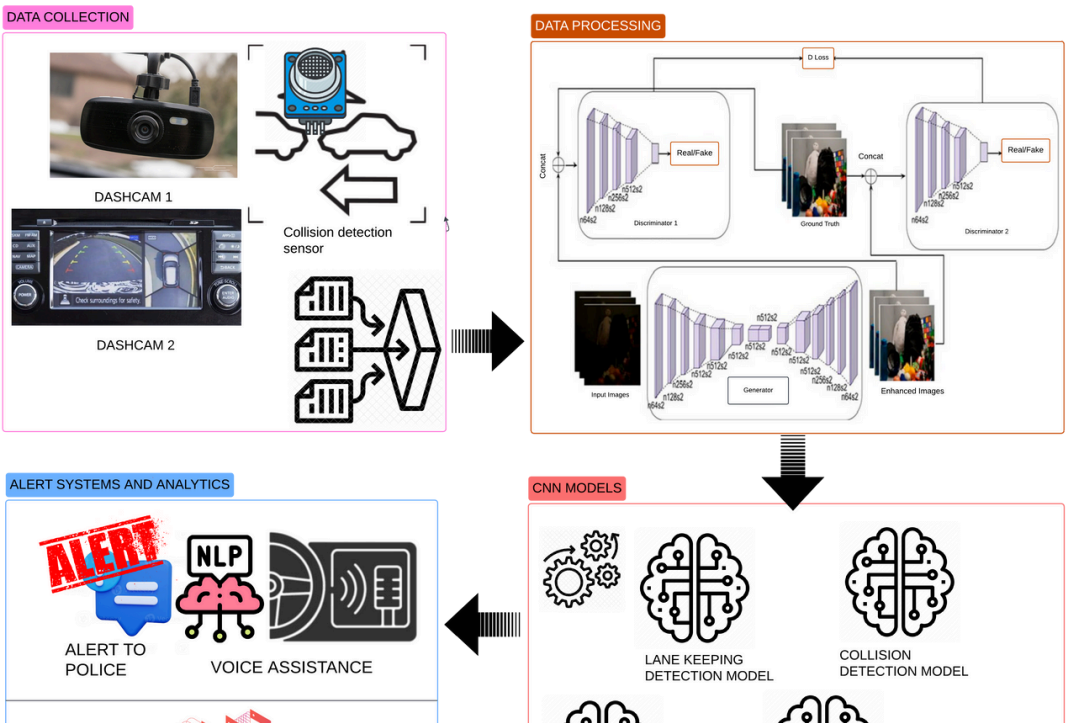
Objectives:

- To detect driver drowsiness and lane switching, and providing alerts before potential accidentsoccurs using CNN, AAQI, OpenCV.
- To detect collision , potholes and make a record of the incident using CNN, Geospatial AI, GIS Integration, OpenCV,.
- To analyze driver behavior and safety system performance using Power BI by visualizing data related to drowsiness detection, pothole and lane-keeping events, and collision incidents.
- To develop a voice assistant that provides real-time safety alerts and notifications for drivers based on detected condition of drowsiness , pothole ,collision using NLP .

Results:



Work Flow:



Key Reference:

- [1] J Gowthamy, A Senthilselvi, Aniket Kumar, S Aakash, and Gandikota Sreedhar. Enhanced ai voice assistance using machine learning and nlp. In 2023 Third International Conference on Smart Technologies, Communication and Robotics (STCR), volume 1, pages 1–5. IEEE,2023.
- [2] R Kannan, Palamakula Jahnvi, and M Megha. Driver drowsiness detection and alert system. In 2023 IEEE International Conference Integrated Circuits and Communication Systems(ICICACS),pages15.IEEE, 2023.
- [3] Vineet Kaushik and Birinderjit Singh Kalyan. Pothole detection system: A review of different methods used for detection. In 2022 Second

Conclusion:

The AI-based Driver Assistance System enhances road safety by providing real-time alerts and proactive interventions, reducing accidents and improving driving conditions.